PROJECT REPORT

Simple To-Do List Manager using Python

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Branch - Integrated MTECH AI

Course - Introduction to Problem Solving and Programing

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Date - 31/03/2025

Introduction

In today’s fast-paced life, managing daily tasks efficiently has become very important, especially for students. Many times, students forget important assignments, deadlines, or personal tasks due to lack of proper organization. I personally noticed that keeping everything in mind is difficult and often leads to confusion and missed work.

To address this issue, I developed a simple To-Do List Manager using Python. The purpose of this project is to provide a basic system where users can easily manage their tasks in an organized manner. The program allows users to add tasks, view them, and delete them once completed.

This project is designed using basic Python concepts, making it easy to understand and use. It is a small but practical application that demonstrates how programming can be applied to solve real-life problems. Even though it is simple, it serves as a foundation for building more advanced task management systems in the future

Objective

The main objective of this project is to develop a simple and user-friendly To-Do List application using Python. The goal is to help users manage their daily tasks efficiently without using complex tools or software.

This project aims to apply the basic programming concepts learned in the course, such as lists, functions, loops, and conditional statements, in a real-world scenario. It also focuses on building logical thinking and problem-solving skills.

Another important objective is to create a menu-driven program that allows users to interact easily with the system. The application should be simple enough for beginners to understand while still being useful in everyday life.

Additionally, this project helps in understanding how to break down a problem into smaller parts and implement each part using functions. It also provides hands-on experience in writing clean and structured code. Overall, the objective is not just to create a program, but to learn how to design and implement a practical solution.

Problem Statement

In daily life, many students and individuals face difficulty in managing their tasks effectively. They often forget important activities such as assignments, meetings, or personal work. This happens because tasks are not properly organized or recorded in a structured way.

Without a proper system, people tend to rely on memory, which is not always reliable. This leads to missed deadlines, increased stress, and reduced productivity. There is a need for a simple and accessible solution that helps users keep track of their daily activities.

The problem becomes more significant for students who have multiple subjects and assignments to manage. A small mistake in remembering tasks can affect their performance. Therefore, a basic task management system is required that is easy to use and does not require any technical knowledge.

This project aims to solve this problem by providing a simple To-Do List application where users can manage their tasks efficiently using a command-line interface.

Methodology

The project is developed using Python programming language, following a simple and structured approach. The main idea is to create a menu-driven program that interacts with the user through the console.

A list data structure is used to store tasks dynamically. Functions are created for each operation such as adding, viewing, and deleting tasks. This makes the code modular and easier to understand.

The program uses a while loop to continuously display the menu options until the user chooses to exit. Based on the user's input, the program performs the corresponding operation. Conditional statements are used to handle different user choices.

The development process involved writing the logic step-by-step, testing each function individually, and then combining them into a complete program. Errors were identified and corrected through repeated testing.

This methodology ensures that the program is simple, efficient, and easy to maintain. It also helps in understanding how basic programming concepts can be combined to build a functional application.

Implementation

The implementation of the To-Do List Manager is based on dividing the program into multiple functions, each responsible for a specific task. The main functions include adding a task, displaying all tasks, and deleting a task.

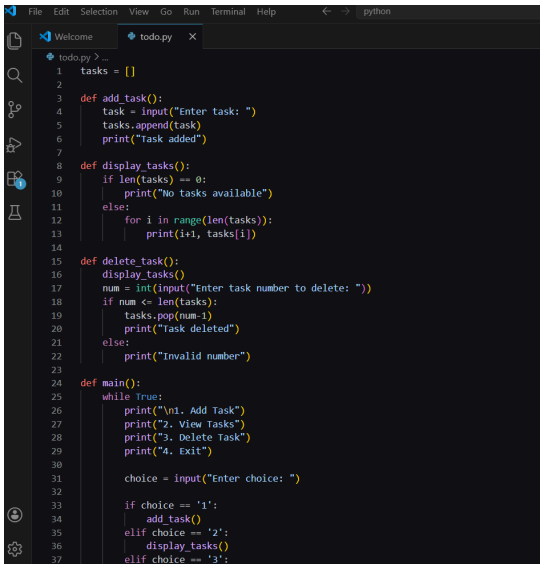
The add_task() function takes input from the user and stores it in a list. The display_tasks() function shows all the tasks along with their serial numbers, making it easier for users to identify them. The delete_task() function allows users to remove a specific task by entering its number.

A menu-driven interface is implemented using a while loop, which keeps the program running until the user selects the exit option. This ensures continuous interaction between the user and the system.

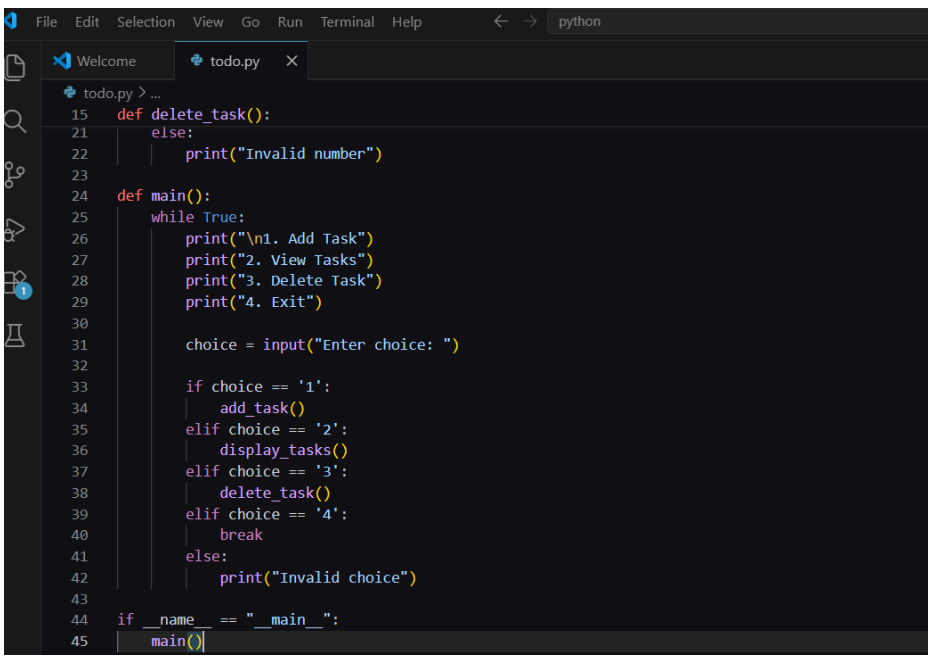
The program uses basic Python concepts such as lists, loops, and conditional statements. The code is written in a simple and readable format to make it easy for beginners to understand.

Overall, the implementation focuses on simplicity, clarity, and functionality, ensuring that the program works efficiently.

INPUT CODE SCREENSHOT



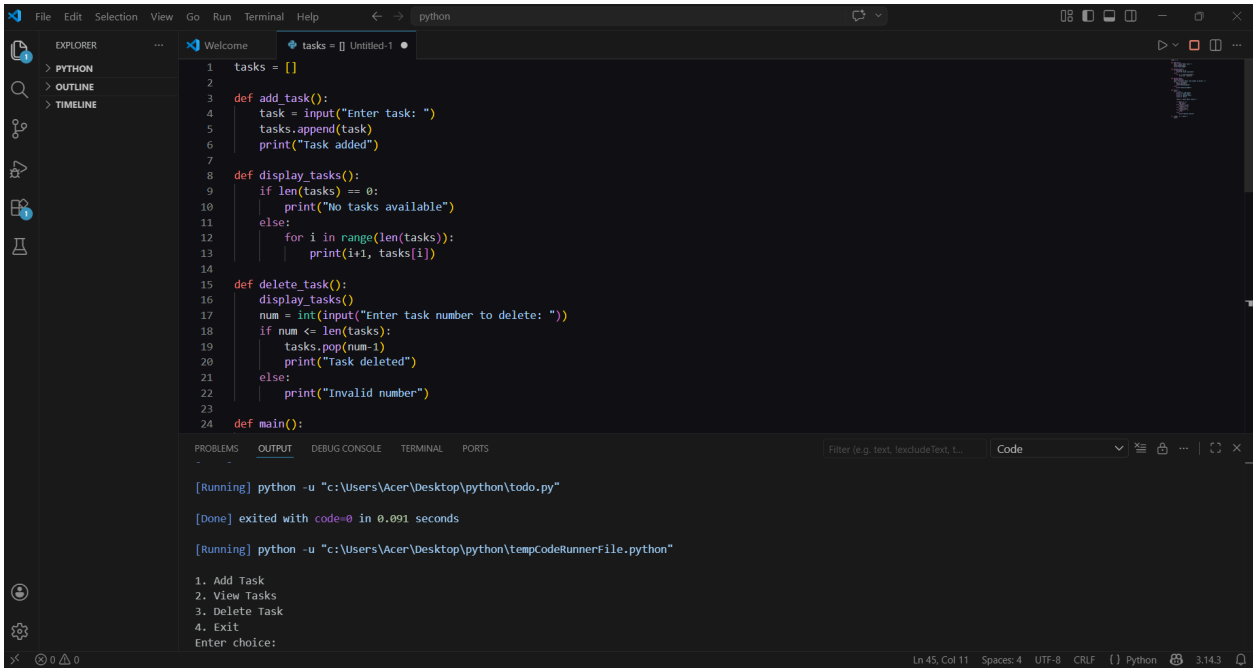
```python
tasks = []

def add_task():
    task = input("Enter task: ")
    tasks.append(task)
    print("Task added")

def display_tasks():
    if len(tasks) == 0:
        print("No tasks available")
    else:
        for i in range(len(tasks)):
            print(i+1, tasks[i])

def delete_task():
    display_tasks()
    num = int(input("Enter task number to delete: "))
    if num <= len(tasks):
        tasks.pop(num-1)
        print("Task deleted")
    else:
        print("Invalid number")

def main():
    while True:
        print("\n1. Add Task")
        print("2. View Tasks")
        print("3. Delete Task")
        print("4. Exit")

        choice = input("Enter choice: ")

        if choice == '1':
            add_task()
        elif choice == '2':
            display_tasks()
```



```python
def delete_task():
    else:
        print("Invalid number")

def main():
    while True:
        print("\n1. Add Task")
        print("2. View Tasks")
        print("3. Delete Task")
        print("4. Exit")

        choice = input("Enter choice: ")

        if choice == '1':
            add_task()
        elif choice == '2':
            display_tasks()
        elif choice == '3':
            delete_task()
        elif choice == '4':
            break
        else:
            print("Invalid choice")

if __name__ == "__main__":
    main()
```

Figure 1: Python code for To-Do List Manager

OUTPUT SCREENSHOT



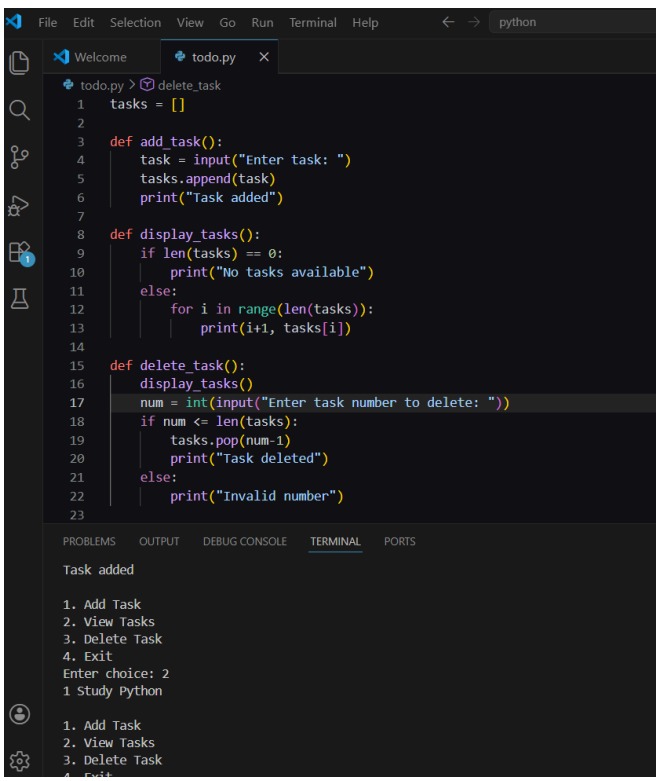


Figure 2: Output showing adding and displaying tasks

Challenges Faced

During the development of this project, several challenges were encountered. One of the main challenges was handling user input correctly. Since users can enter invalid data, it was important to ensure that the program does not crash and handles errors properly.

Another challenge was managing task numbers while deleting tasks. Since the list changes dynamically, maintaining the correct indexing required careful handling.

Designing a clear and simple menu system was also slightly challenging. The program needed to be easy to use, so the menu had to be structured in a way that users could understand without confusion.

Additionally, debugging the program and identifying logical errors required patience and repeated testing. Small mistakes in indentation or conditions could affect the program’s behavior.

These challenges were overcome by testing the program multiple times, analyzing errors, and improving the logic step by step.

Learning Outcomes

This project provided valuable learning experience in Python programming. It helped in understanding how basic concepts such as lists, loops, and functions can be applied to build a real-world application.

I learned how to structure a program using functions, which makes the code more organized and easier to manage. It also improved my understanding of user input handling and conditional logic.

The project enhanced my problem-solving skills, as I had to think step-by-step to design the logic of the program. It also taught me the importance of testing and debugging in programming.

Working on this project gave me confidence in writing Python programs independently. I also understood how small applications can solve practical problems effectively.

Conclusion

The To-Do List Manager is a simple yet useful application that helps users manage their daily tasks efficiently.
Although the project is basic, it demonstrates how programming can be used to solve real-life problems. With further improvements, this project can be expanded into a more advanced task management system.

Future Enhancements

This project can be improved in many ways to make it more advanced and user-friendly. One of the major improvements would be adding a feature to save tasks in a file so that data is not lost after the program is closed.Another enhancement could be adding a task completion status, which allows users to mark tasks as completed instead of deleting them. This would help in better task tracking.A graphical user interface (GUI) can also be developed using libraries like Tkinter, making the application more interactive and visually appealing.Additional features such as reminders, deadlines, and priority levels can be included to make the application more powerful.

These enhancements would transform the project from a basic program into a more professional and practical task management system.